The `last!` method behaves exactly like `last`, except that it will raise `ActiveRecord::RecordNotFound` if no matching record is found.

1.1.5 `find_by`

The `find_by` method finds the first record matching some conditions. For example:

```
Client.find_by first_name: 'Lifo'
# => #<Client id: 1, first_name: "Lifo">

Client.find_by first_name: 'Jon'
# => nil
```

It is equivalent to writing:

```
Client.where(first_name: 'Lifo').take
```

The SQL equivalent of the above is:

```
SELECT * FROM clients WHERE (clients.first_name = 'Lifo') LIMIT 1
```

The `find_by!` method behaves exactly like `find_by`, except that it will raise `ActiveRecord::RecordNotFound` if no matching record is found. For example:

```
Client.find_by! first_name: 'does not exist'
# => ActiveRecord::RecordNotFound
```

This is equivalent to writing:

```
Client.where(first_name: 'does not exist').take!
```

1.2 Retrieving Multiple Objects in Batches

We often need to iterate over a large set of records, as when we send a newsletter to a large set of users, or when we export data.

This may appear straightforward:

```
# This is very inefficient when the users table has thousands of rows.
User.all.each do |user|
  NewsMailer.weekly(user).deliver_now
end
```

But this approach becomes increasingly impractical as the table size increases, since `User.all.each` instructs Active Record to fetch *the entire table* in a single pass, build a model object per row, and then keep the entire array of model objects in memory. Indeed, if we have a large number of records, the entire collection may exceed the amount of memory available.

Rails provides two methods that address this problem by dividing records into memory-friendly batches for processing. The first method, `find_each`, retrieves a batch of records and then yields*each* record to the block individually as a model. The second method, `find_in_batches`, retrieves a batch of records and then yields *the entire batch* to the block as an array of models.

The `find_each` and `find_in_batches` methods are intended for use in the batch processing of a large number of records that wouldn't fit in memory all at once. If you just need to loop over a thousand records the regular find methods are the preferred option.

1.2.1 `find_each`

The `find_each` method retrieves a batch of records and then yields *each* record to the block individually as a model. In the following example, `find_each` will retrieve 1000 records (the current default for both `find_each` and `find_in_batches`) and then yield each record individually to the block as a model. This process is repeated until all of the records have been processed:

```
User.find_each do |user|
```